

BUDT704_Data Processing and Analysis in Python

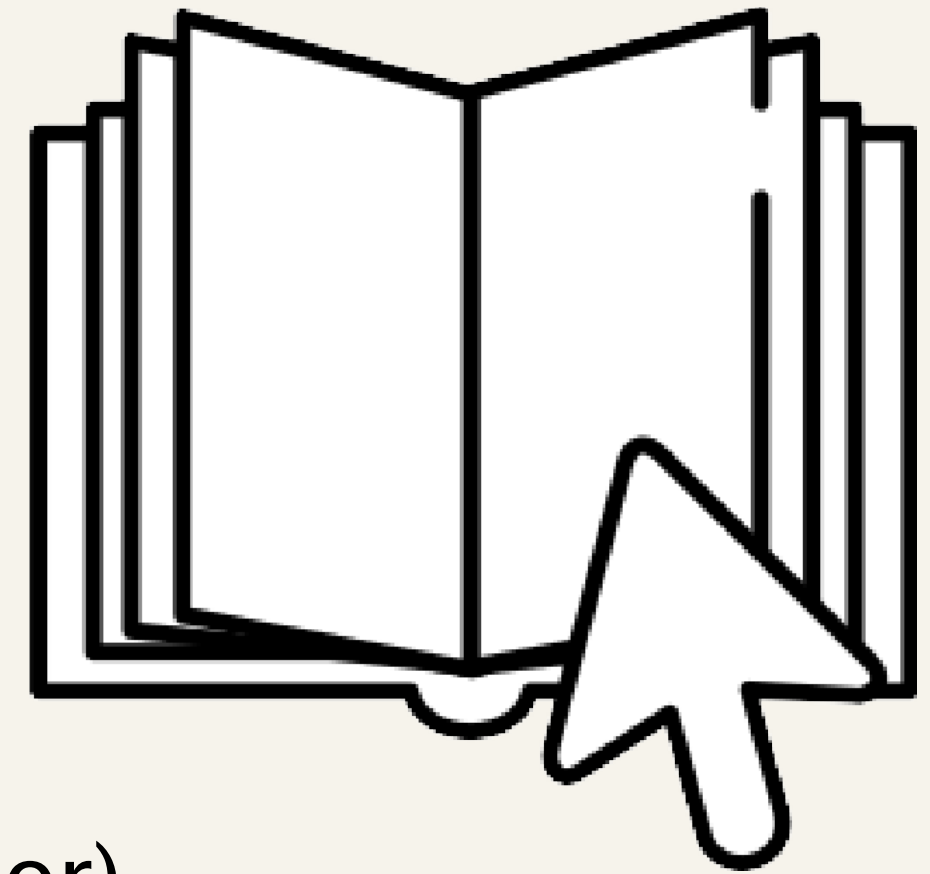
Predicting Hotel Booking Cancellations

**Presented By :
Prasham Shah**



INDEX

- Problem Statement
- Objectives
- Exploratory Data Analysis
- Feature Engineering
- Backward Elimination
- Logistic Regression Model
- Decision Tree Regressor
- Ensemble Model (Bagging, Boosting, Random Forest Regressor)
- Classifier Models
- Ensemble Model of Classifier Models
- Variable Importance Plot
- Conclusion



PROBLEM STATEMENT

The project aims to develop a predictive model to identify hotel booking cancellations, a critical task to mitigate revenue loss and optimize operations. It involves preprocessing booking data (e.g., lead time, deposit type, special requests) through feature selection, handling missing values, and reducing noise. Multiple models will be trained and evaluated to determine the most effective approach, with an analysis of key features influencing cancellations.

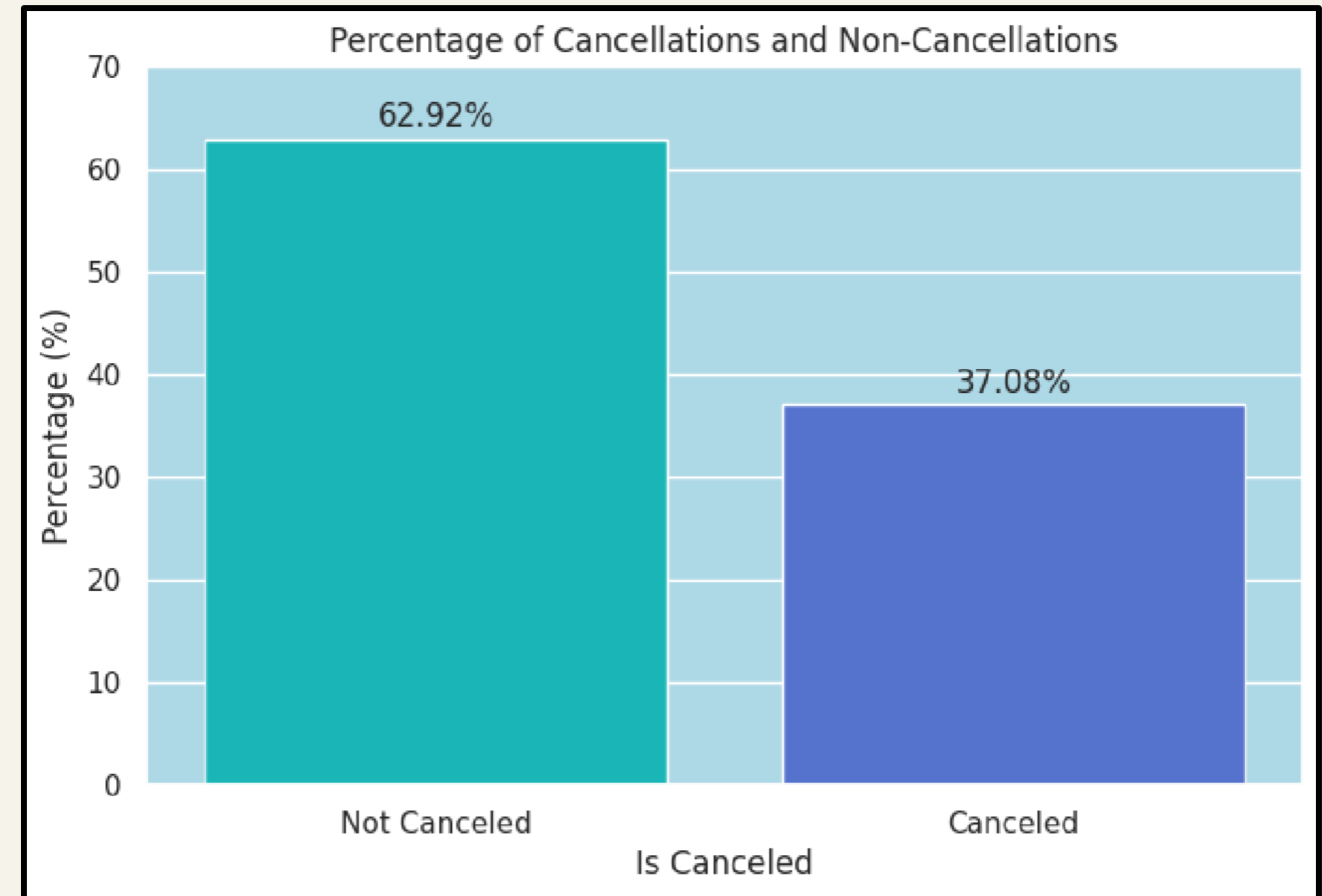
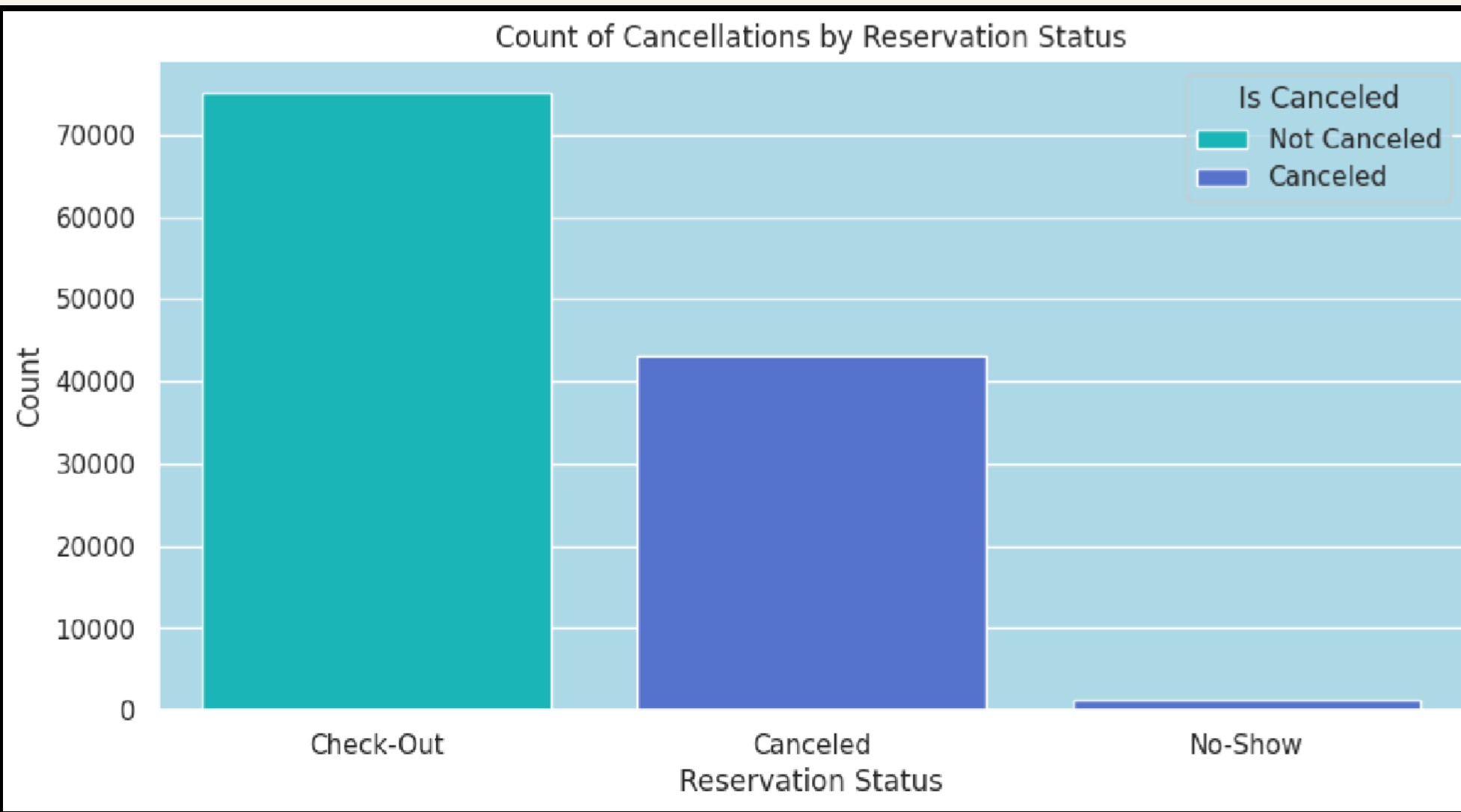




OVERVIEW

- **Dataset Exploration:** Analyze dataset structure with summary statistics for numerical and categorical variables.
- **Data Preprocessing:**
 - **Select and engineer key features.**
 - **Handle missing and noisy data.**
 - **Encode categorical variables for modeling.**
- **Model Development:** Train and optimize Decision Trees, Random Forest, and XGBoost, focusing on maximizing F1-score for class 1.
- **Model Evaluation:** Compare models using accuracy, precision, recall, F1-score, and AUC.
- **Feature Importance:** Analyze key features influencing model predictions.

EXPLANATORY DATA ANALYSIS



FEATURE ENGINEERING



1

Created new features like interaction terms, time-based differences, and polynomial features to capture complex relationships in the data.

2

Applied encoding techniques i.e. one hot encoding for categorical variables and scaling/normalization for numerical features to optimize model performance.

3

Managed missing values using imputation strategies and filtered out noisy or irrelevant features to enhance data quality and predictive power.

LOGISTIC REGRESSION MODEL

Logistic Regression: Logistic regression is a statistical model used for binary classification tasks. It predicts the probability of a binary outcome (e.g., 0 or 1) by modeling the relationship between input features and the target variable using a logistic function (sigmoid). The output is a value between 0 and 1, which is interpreted as the likelihood of a specific class.

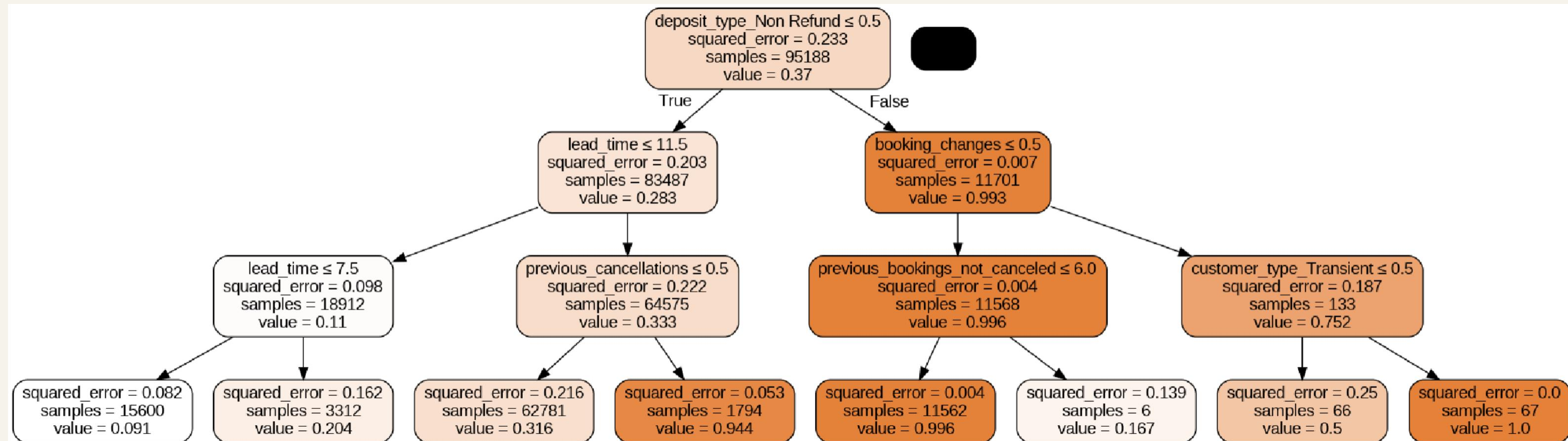
Accuracy of Logistic Regression: 0.788376686136908					
	precision	recall	f1-score	support	
0	0.79	0.90	0.84	14917	
1	0.78	0.61	0.68	8880	
accuracy			0.79	23797	
macro avg	0.79	0.75	0.76	23797	
weighted avg	0.79	0.79	0.78	23797	

DECISION TREE REGRESSOR

Prediction on Training Data: 0.404

Prediction on Test Data: 0.407

K-Fold(= 3) = 0.405



ENSEMBLE MODEL

Bagging

Random Forest Regressor

Boosting

How good is the bagging model if subjected to variations in the training sample that is the test data?

0.3207313246587909

How good is this prediction of bagging in training?

Regression statistics

Mean Error (ME) : -0.0020
Root Mean Squared Error (RMSE) : 0.1282
Mean Absolute Error (MAE) : 0.0777

How good is this prediction of random forest in training?

Regression statistics

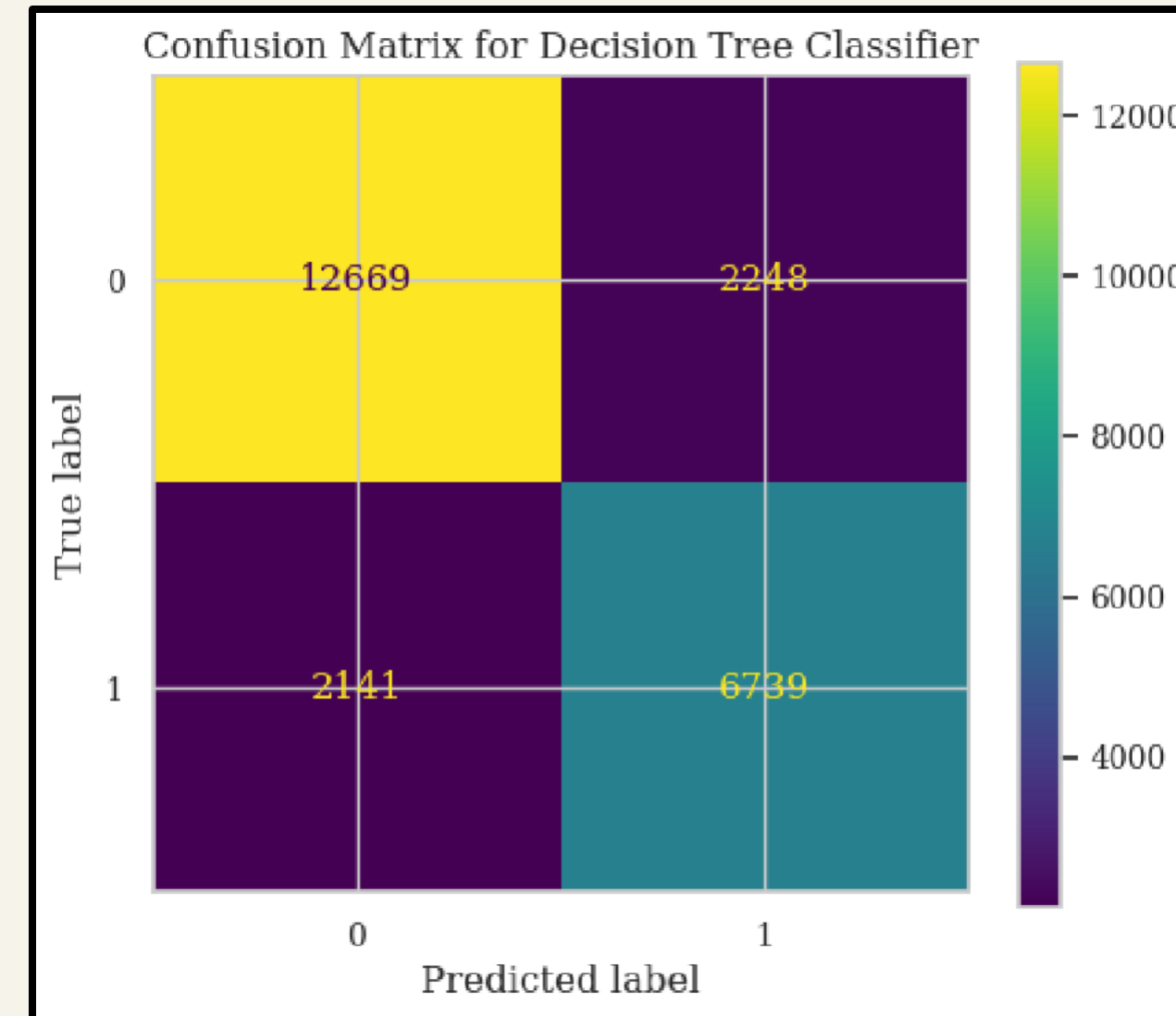
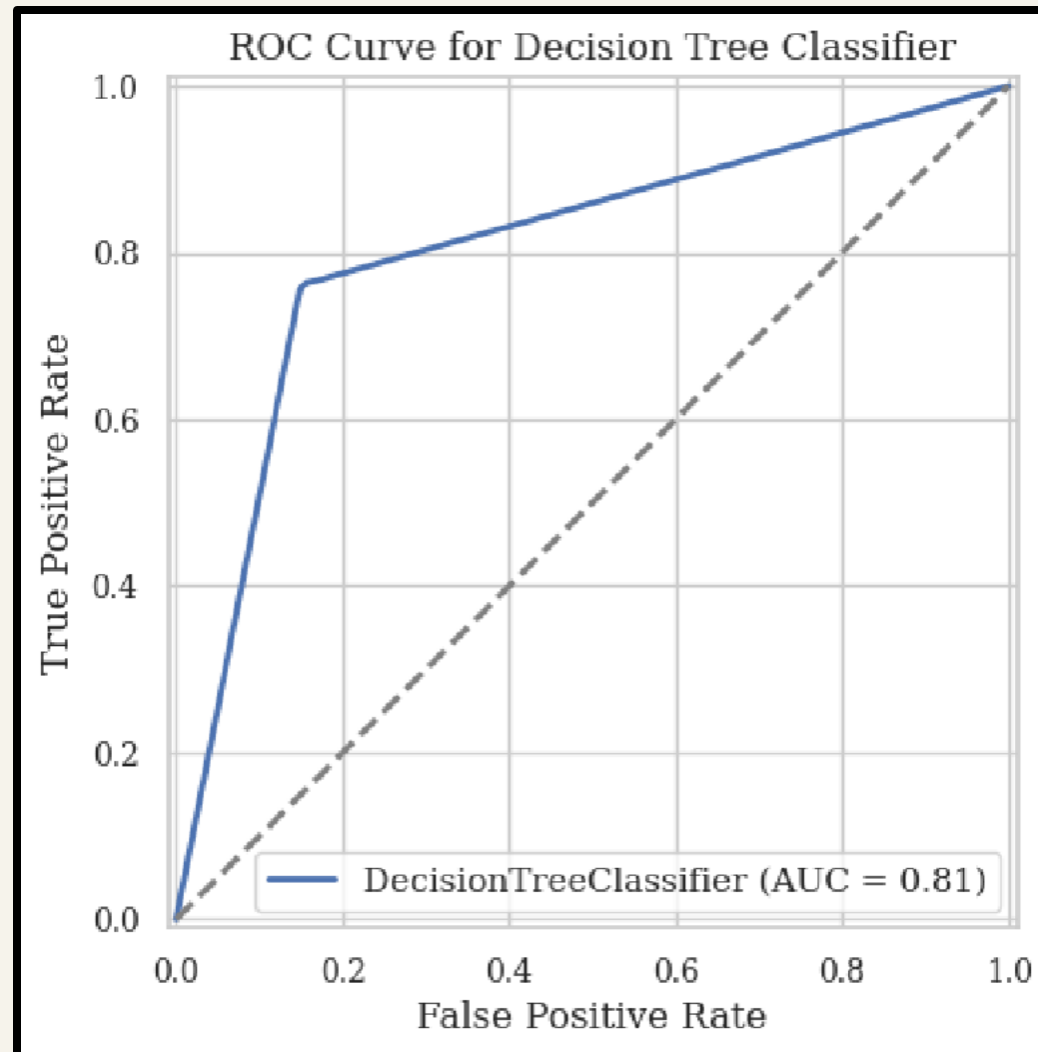
Mean Error (ME) : -0.0019
Root Mean Squared Error (RMSE) : 0.1283
Mean Absolute Error (MAE) : 0.0780

How good is this prediction of random forest in training?

Regression statistics

Mean Error (ME) : 0.0000
Root Mean Squared Error (RMSE) : 0.3652
Mean Absolute Error (MAE) : 0.2869

DECISION TREE CLASSIFIER



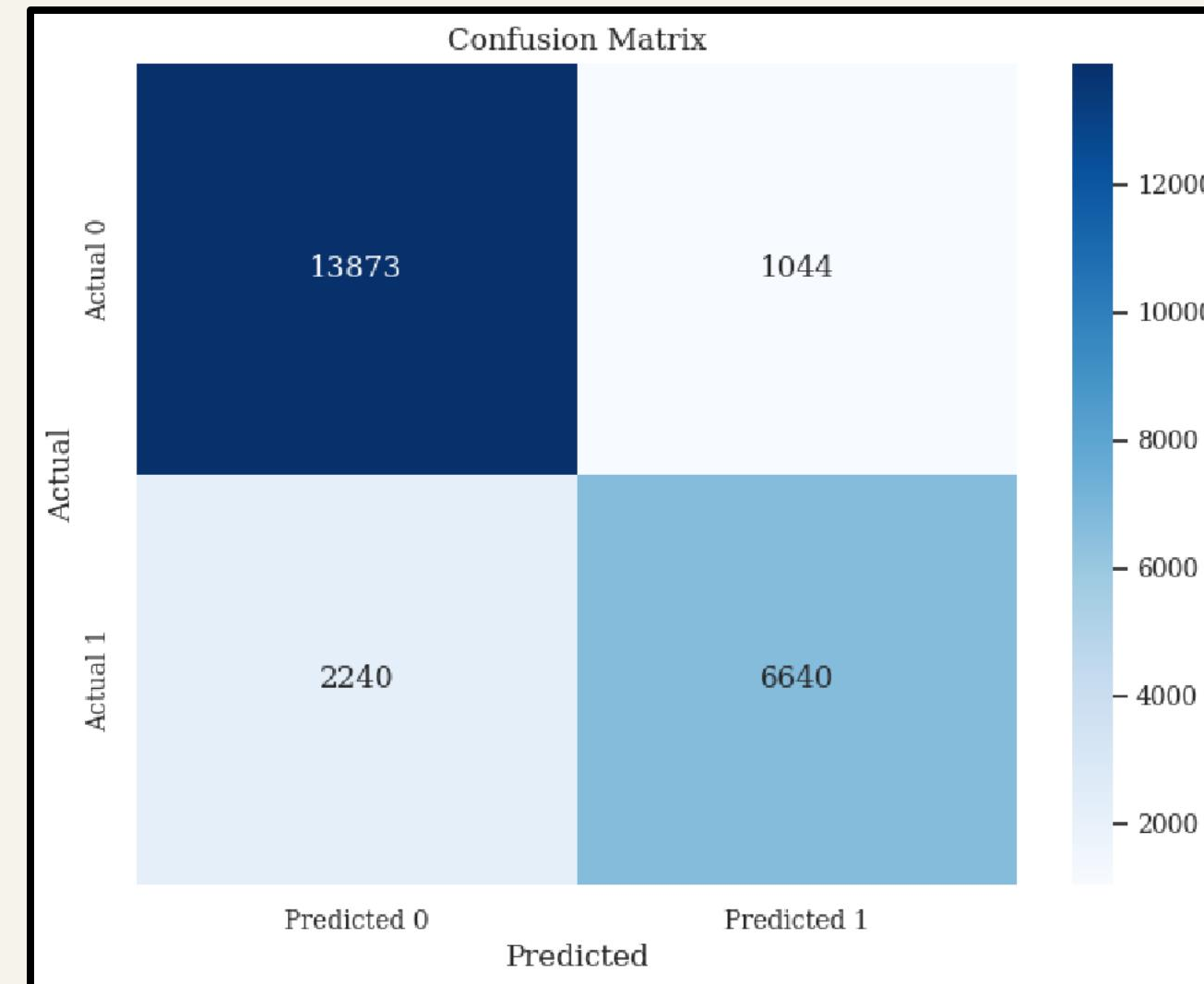
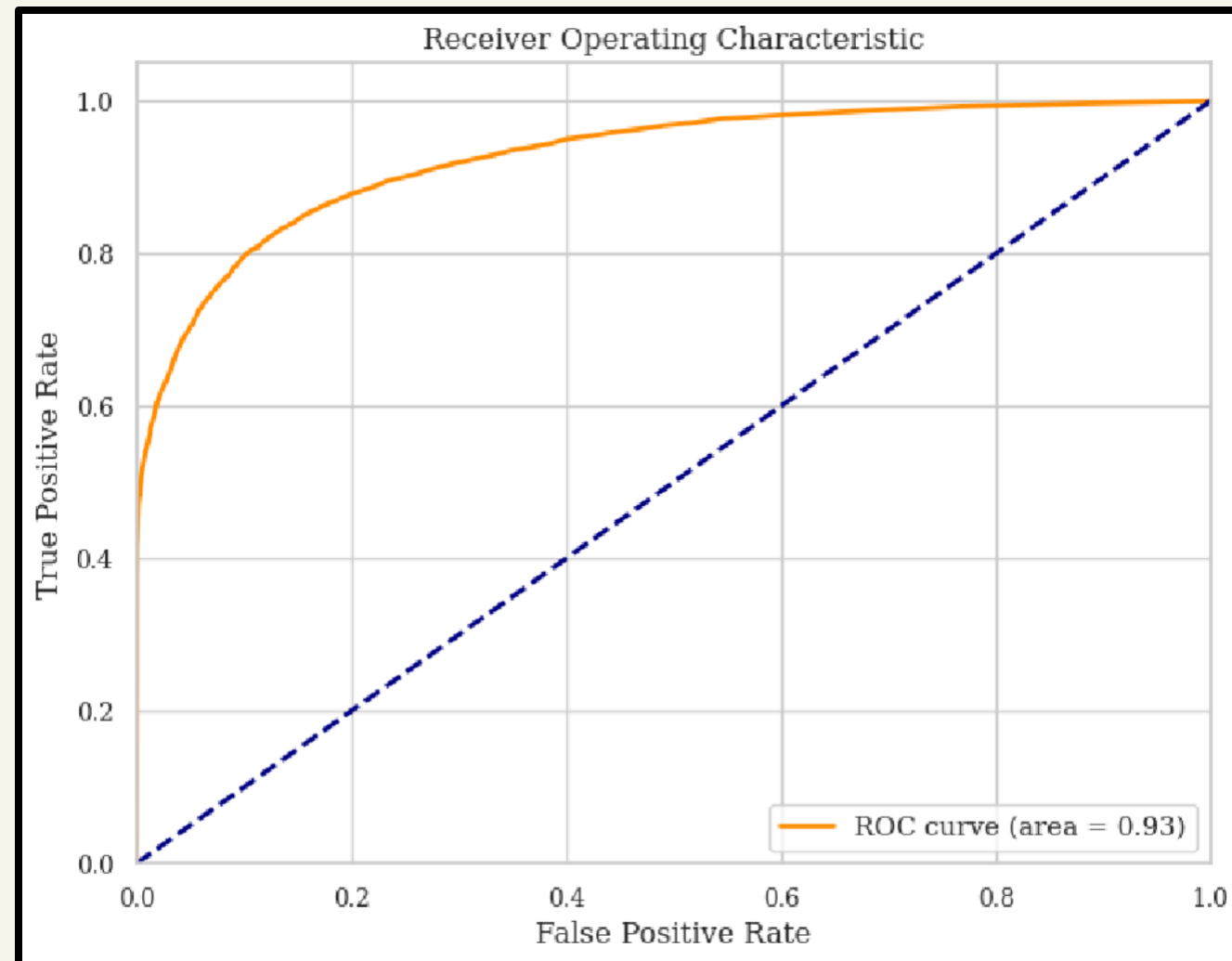
```
Decision Tree Classifier Evaluation:
      precision    recall  f1-score   support

     0       0.86      0.85      0.85     14917
     1       0.75      0.76      0.75      8880

 accuracy          0.82     23797
 macro avg         0.80      0.80      0.80     23797
 weighted avg      0.82      0.82      0.82     23797

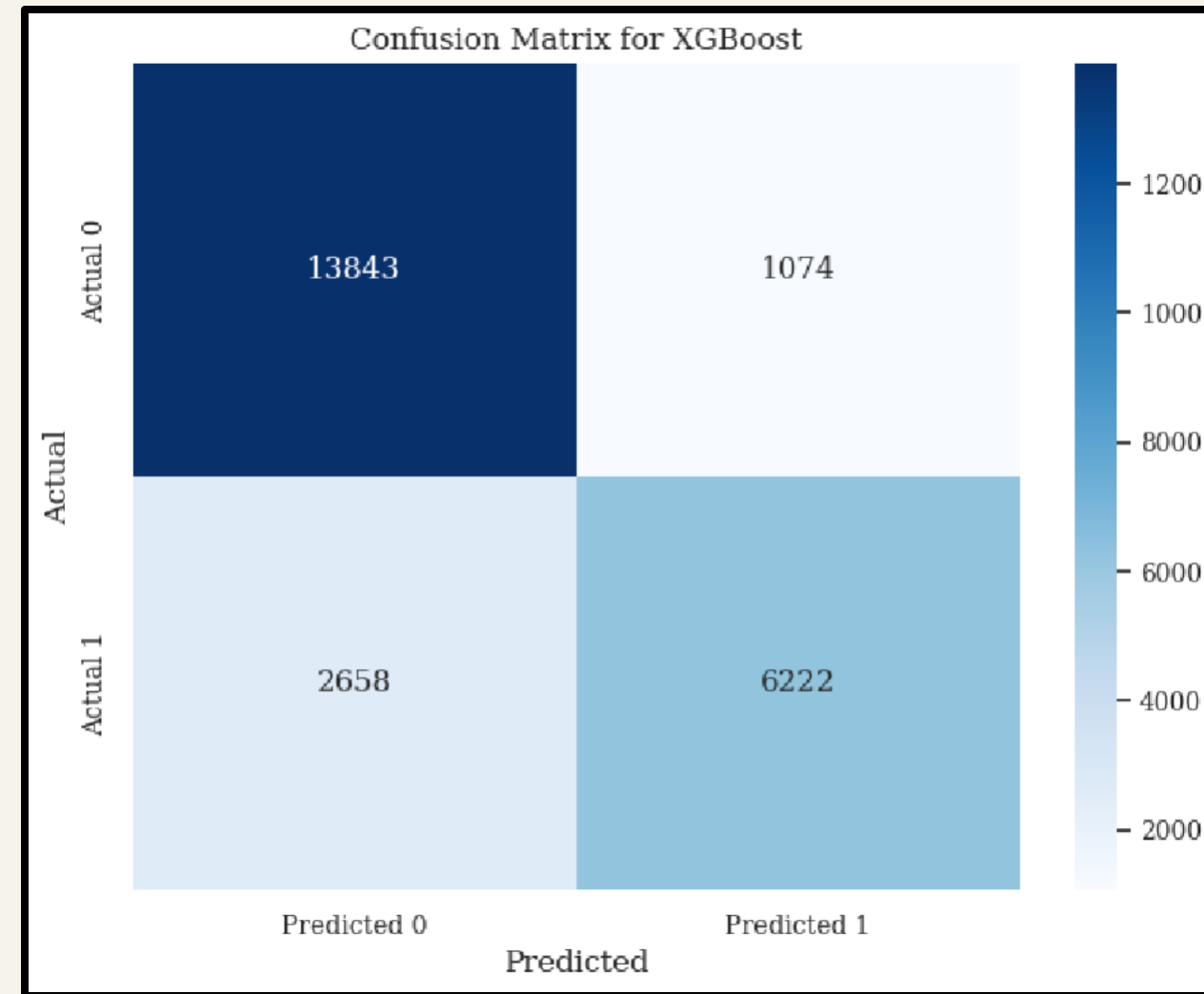
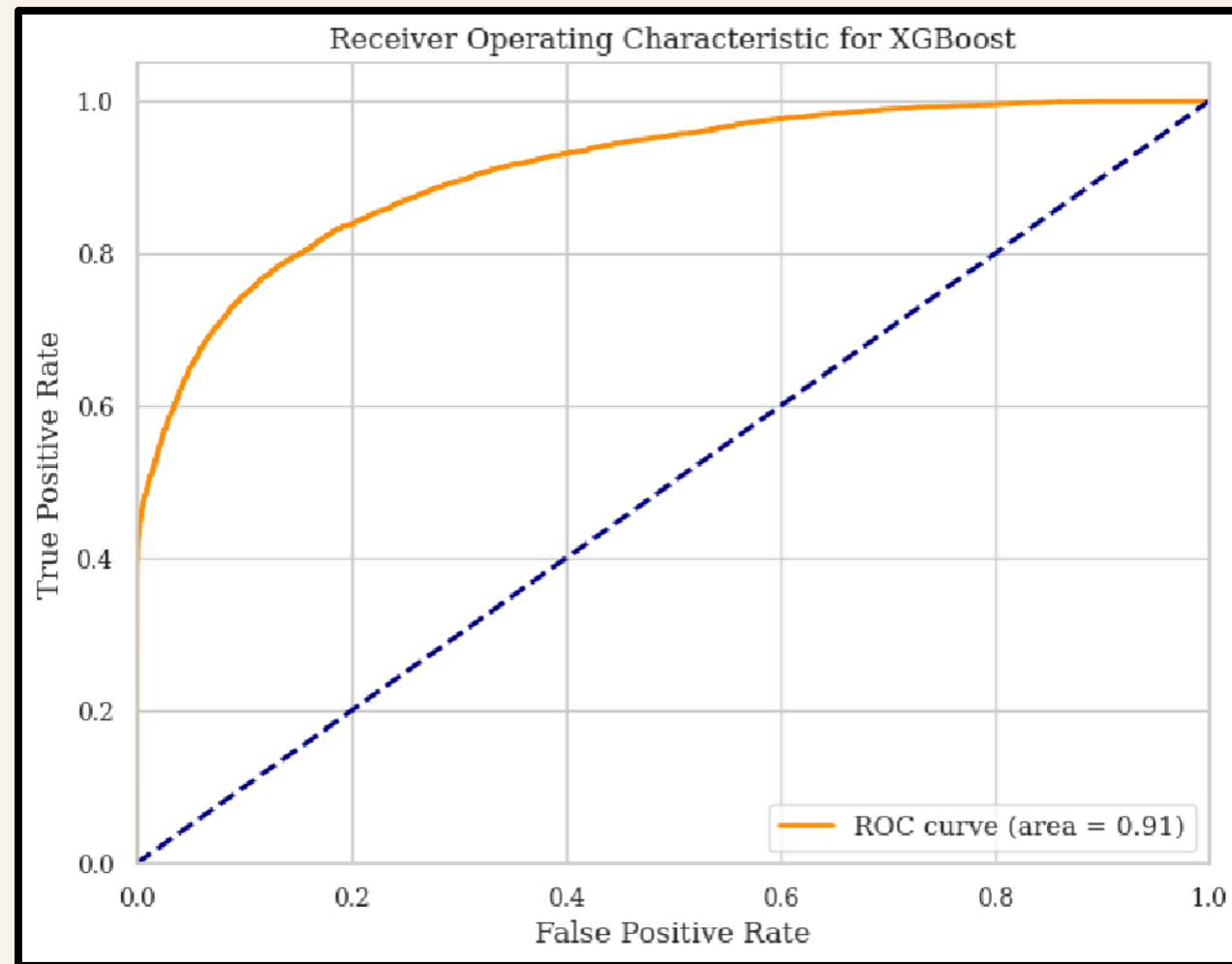
Accuracy: 0.8155649871832584
ROC AUC: 0.8057925853385732
```

RANDOM FOREST CLASSIFIER



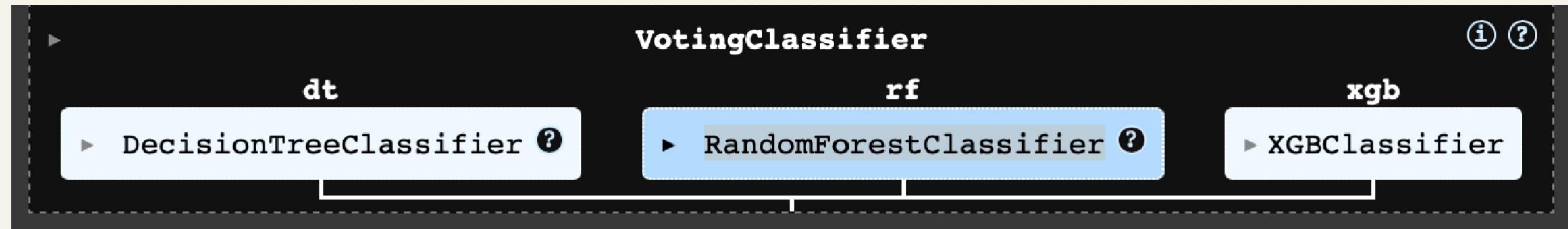
	precision	recall	f1-score	support
0	0.86	0.93	0.89	14917
1	0.86	0.75	0.80	8880
accuracy			0.86	23797
macro avg	0.86	0.84	0.85	23797
weighted avg	0.86	0.86	0.86	23797

XGBOOST CLASSIFIER



	precision	recall	f1-score	support
0	0.84	0.93	0.88	14917
1	0.85	0.70	0.77	8880
accuracy			0.84	23797
macro avg	0.85	0.81	0.83	23797
weighted avg	0.84	0.84	0.84	23797

ENSEMBLE OF CLASSIFIER MODELS



```
Ensemble Model Evaluation:
      precision    recall  f1-score   support

     0       0.86      0.90      0.88     14917
     1       0.81      0.75      0.78      8880

 accuracy          0.84      23797
 macro avg         0.84      0.83      0.83      23797
weighted avg         0.84      0.84      0.84      23797

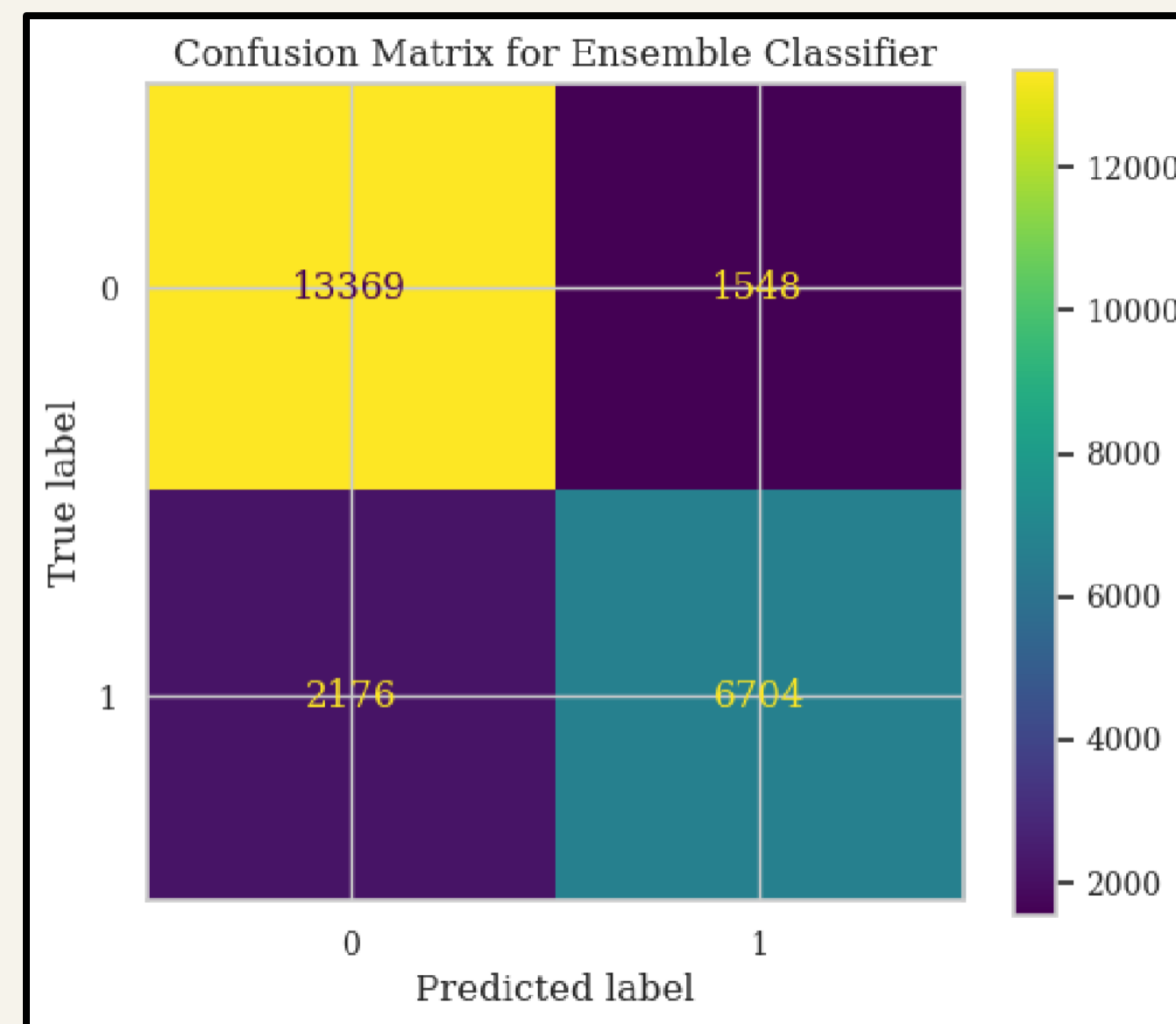
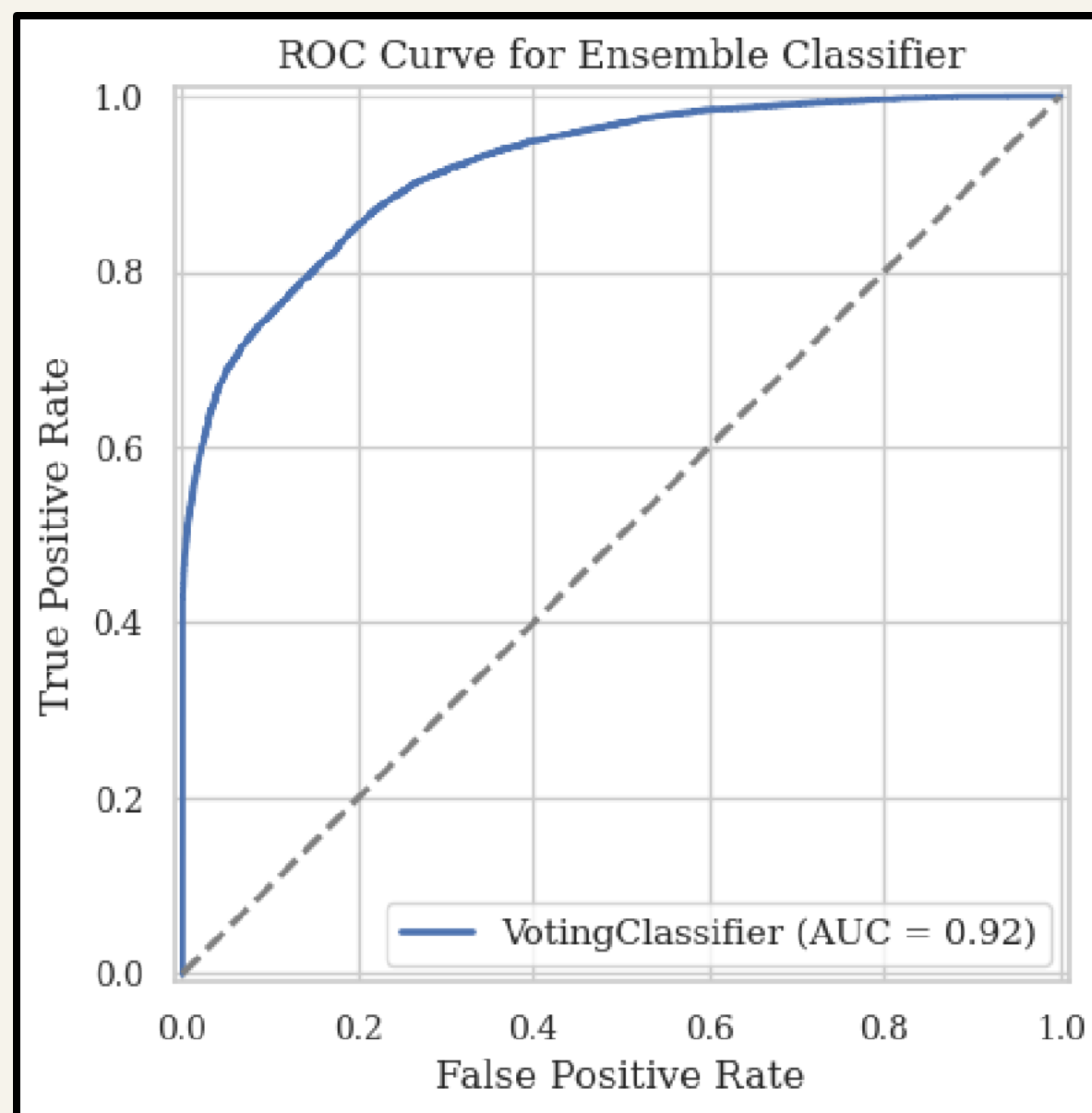
Accuracy: 0.8435096860948859
ROC AUC: 0.9201195413419722
```


ENSEMBLE OF CLASSIFIER MODELS

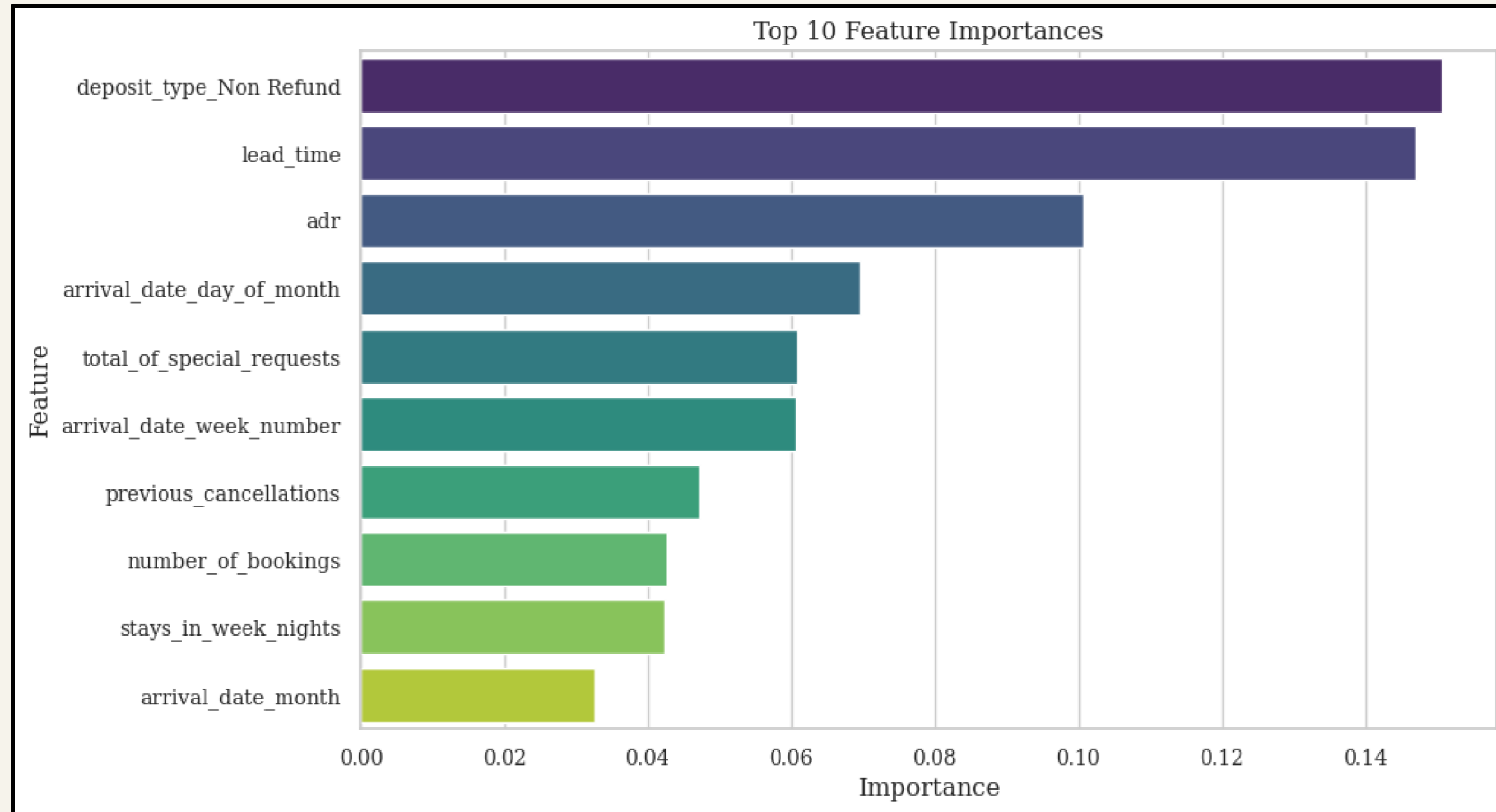
Decision Tree Classifier

Random Forest Classifier

XG Boost Classifier



FEATURE IMPORTANCE



TIMELINE



CONCLUSION

1

The Random Forest model emerged as the most effective predictor, with XGBoost delivering competitive results, highlighting their reliability for predicting cancellations.

2

Key factors driving predictions include customer type, lead time, deposit type, and market segment, offering actionable insights for operational strategies.

3

This project equips hotels with a powerful tool to optimize resource allocation, minimize financial losses, and enhance customer satisfaction by predicting cancellations effectively.



The background features three vertical stripes on the left: a wide pink stripe, a medium blue stripe, and a narrow beige stripe. The right side of the slide is a light cream color, decorated with two rectangular areas of small, light pink dots in the top right and bottom right corners.

THANK YOU